

Einstein's General Theory of Relativity

Lecture 8: Curvature from Commutators and the Newtonian Limit

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Chapter 1

Curvature from Commutators and the Newtonian Limit

Curvature is easy to picture and harder to calculate. A vector carried around a small closed loop generally returns with a changed orientation, but turning that geometric statement into a usable formula requires careful notation. We shall introduce only as much operator language as is needed to make the calculation transparent. The reward is substantial: the Riemann tensor will emerge as a commutator, its contractions will lead to the Ricci tensor and scalar curvature, and the same geometric language will then meet Newtonian gravity in the weak-field limit.

The lecture has two distinct movements. The first is mathematical. We derive curvature by comparing covariant differentiation in two orders. The second is physical. We place a slowly moving particle in a weak, nearly static field and ask how the geodesic equation can reproduce $F = ma$ and Poisson's equation. Those two movements meet in the search for a generally covariant field equation.

1.1 Vector Fields and Linear Operators

Let

$$V^\alpha(x) = (V^0(x), V^1(x), V^2(x), V^3(x)) \quad (1.1)$$

be a vector field on four-dimensional spacetime. It is useful to regard its four components as four functions collected into one column vector. Several familiar operations on this collection are linear operators.

First, differentiation with respect to a coordinate is an operator:

$$\partial_\mu V^\alpha(x) = \frac{\partial V^\alpha(x)}{\partial x^\mu}. \quad (1.2)$$

Second, multiplication by an ordinary function is an operator:

$$F(x) V^\alpha(x). \quad (1.3)$$

Third, a matrix acts on the component index:

$$(MV)^\alpha = M^\alpha_\beta V^\beta. \quad (1.4)$$

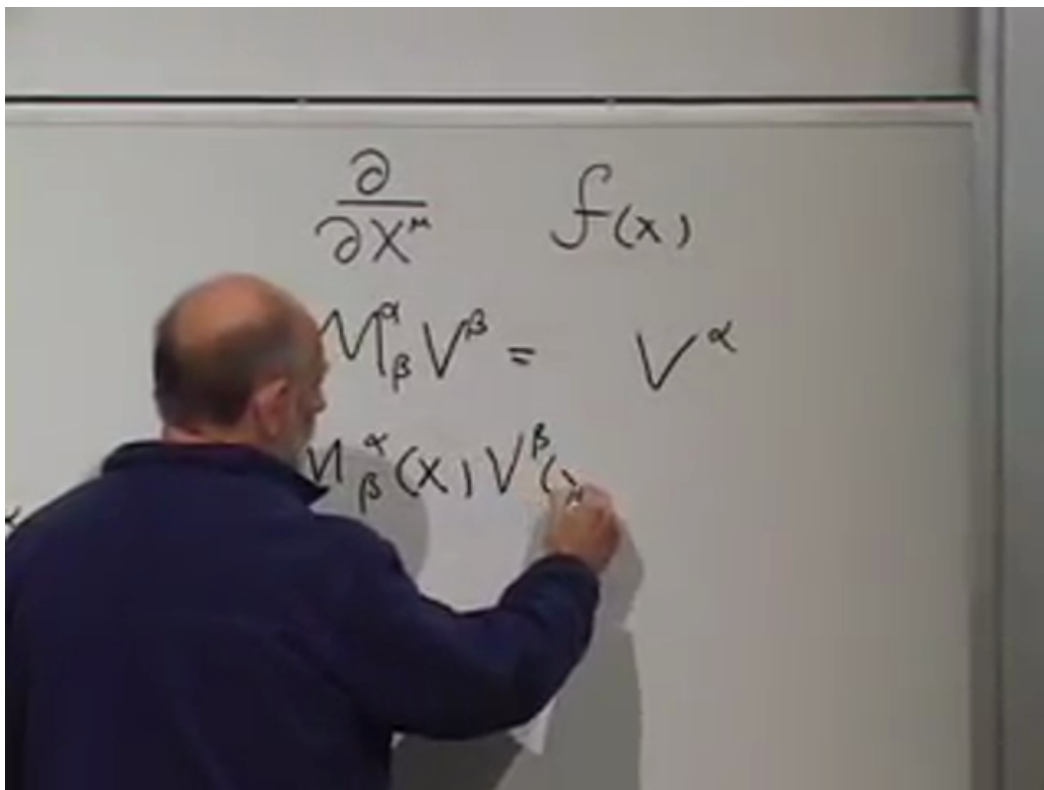


Figure 1.1: Derivative, function-multiplication, and matrix-multiplication operators acting on the component functions of a vector field. The final line allows the matrix itself to depend on position.

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The matrix may itself depend on position, $M^\alpha_\beta = M^\alpha_\beta(x)$. These examples are not an exhaustive catalogue; they are exactly the ingredients needed below.

The covariant derivative combines differentiation with a position-dependent matrix:

$$(\nabla_\mu V)^\alpha = \partial_\mu V^\alpha + \Gamma^\alpha_{\mu\beta} V^\beta. \quad (1.5)$$

For each fixed value of μ , define the matrix

$$(\Gamma_\mu)^\alpha_\beta = \Gamma^\alpha_{\mu\beta}. \quad (1.6)$$

Then the same definition can be written schematically as

$$\boxed{\nabla_\mu = \partial_\mu + \Gamma_\mu}. \quad (1.7)$$

The indices α, β are hidden matrix indices; μ labels which directional derivative and which connection matrix are being used. Whenever the compact notation becomes ambiguous, Eq. (1.5) restores all three roles.

Question from the audience. Why are there four connection matrices? Is that tied to the three spatial coordinates and one time coordinate? ^a

Response. Yes, in this discussion spacetime is four-dimensional, so μ takes four values and labels four matrices Γ_μ . In an n -dimensional manifold there would be n such matrices, each of size $n \times n$. The construction is not tied to four dimensions.

^aLecture timestamp: 00:12:40–00:13:08.

The image shows two equations written on a chalkboard. The top equation is the component definition of the covariant derivative of a vector field V^α along a direction μ :

$$\nabla_\mu V^\alpha = \frac{\partial V^\alpha}{\partial x^\mu} + \Gamma_{\mu\beta}^\alpha V^\beta$$
 The indices α and β are written in red, while μ is black. The bottom equation is the compact operator form:

$$\nabla_\mu = \frac{\partial}{\partial x^\mu} + \Gamma_\mu^\cdot$$
 The dot is written in red.

Figure 1.2: The component definition of the covariant derivative and its compact operator form. The colored indices on the board distinguish the matrix slots from the derivative direction.

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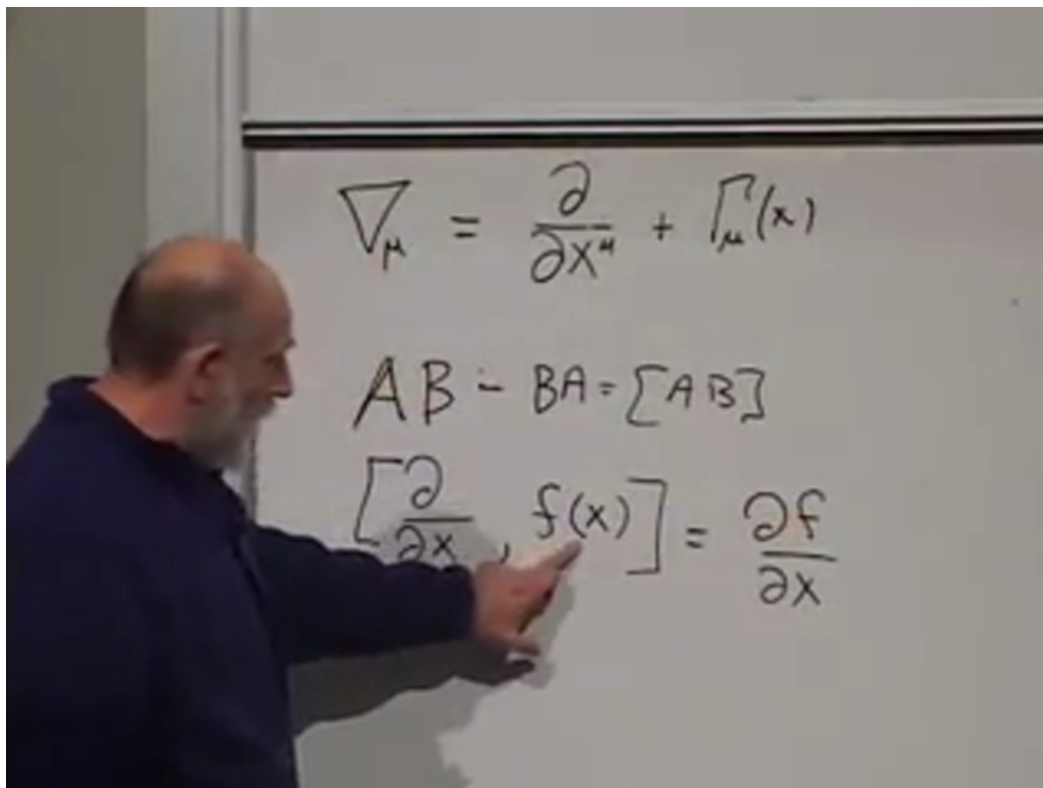


Figure 1.3: The compact covariant-derivative notation, the definition of a commutator, and the derivative–multiplication identity that drives the curvature calculation.

Lecture frame: 00:15:30.

1.2 The One Commutator Identity We Need

For two operators A and B , their product AB means that B acts first and A acts second. Their commutator is

$$[A, B] = AB - BA. \quad (1.8)$$

For noncommuting operators, reversing the sequence can change the result.

Regard $F(x)$ as the operator that multiplies by the function F . The elementary identity needed for curvature is

$$\boxed{[\partial_x, F(x)] = \frac{dF}{dx}.} \quad (1.9)$$

This is an operator identity, so its meaning must be tested on an arbitrary function $V(x)$:

$$\begin{aligned} [\partial_x, F]V &= \partial_x(FV) - F\partial_x V \\ &= \frac{dF}{dx}V + F\frac{dV}{dx} - F\frac{dV}{dx} \\ &= \frac{dF}{dx}V. \end{aligned} \quad (1.10)$$

The terms in which the derivative acts on V cancel. The remaining instruction is simply multiplication by dF/dx .

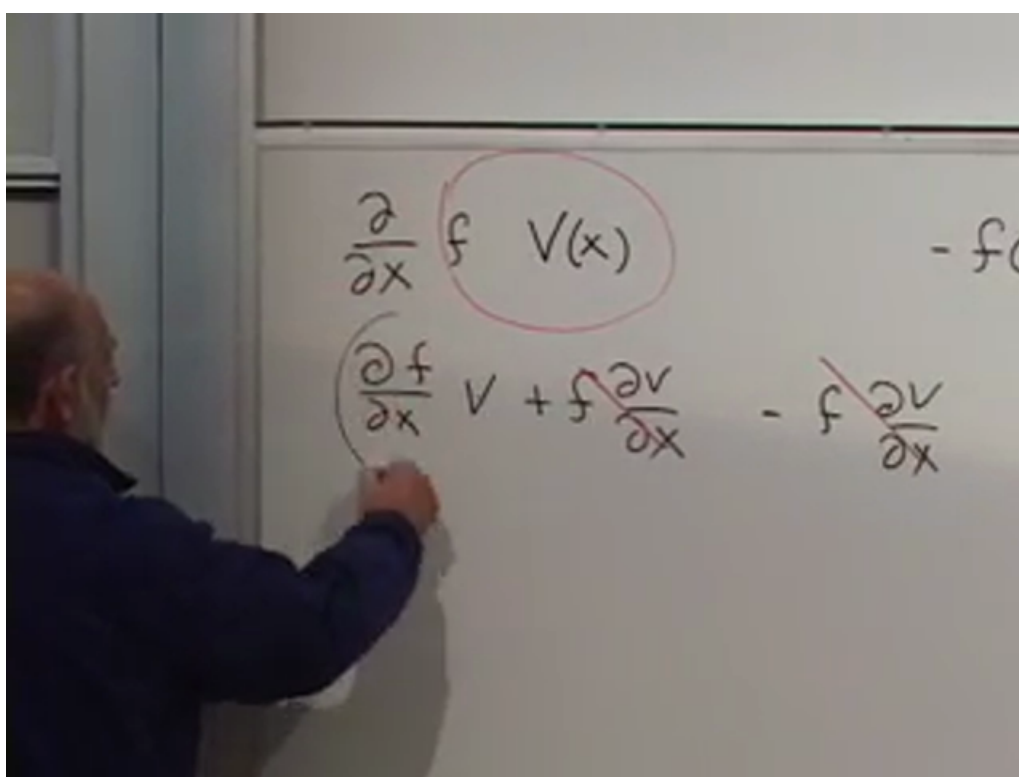


Figure 1.4: The product-rule calculation behind Eq. (1.9). The two terms in which the derivative reaches the test function cancel.

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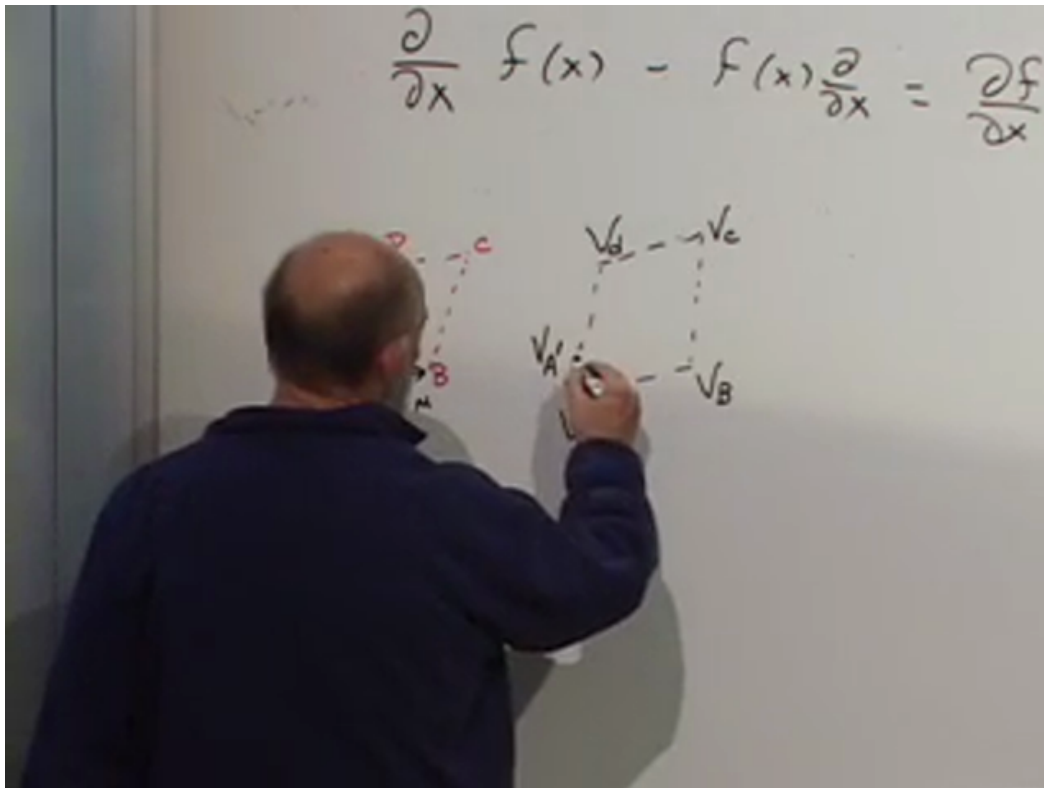


Figure 1.5: The transported vector around an infinitesimal loop. The dashed offset at A records its change in orientation after the path has returned to its starting point.

Lecture frame: 00:23:30.

The same identity holds component by component when F is replaced by a matrix-valued function such as $\Gamma_\mu(x)$:

$$[\partial_\nu, \Gamma_\mu] = \partial_\nu \Gamma_\mu. \quad (1.11)$$

This does not mean that a matrix has become a scalar. It means that every matrix entry is differentiated.

1.3 A Loop as a Difference of Differences

Choose two infinitesimal displacement vectors, ϵ^μ and η^ν , spanning a small parallelogram. Label its corners A, B, C, D , and parallel transport a vector successively along

$$A \longrightarrow B \longrightarrow C \longrightarrow D \longrightarrow A.$$

The base point returns to A , but the transported vector need not return with the same components. Denote the initial vector by V_A and the final one by V'_A .

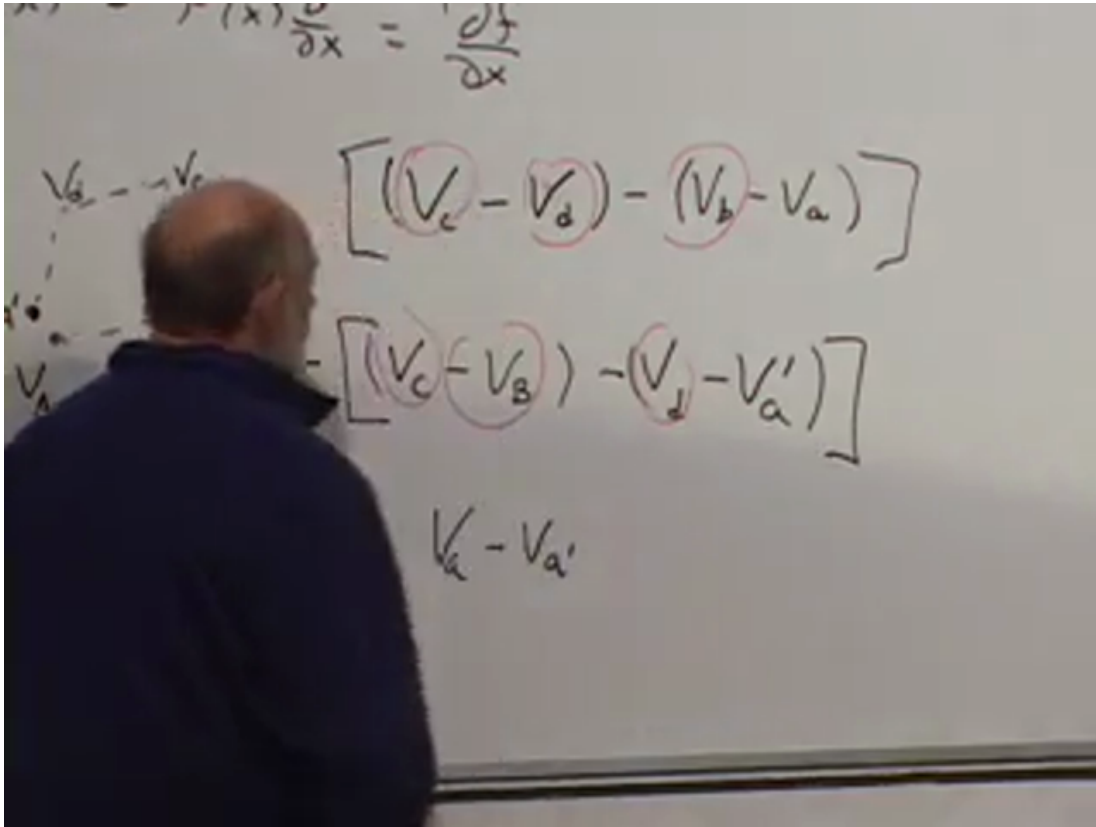


Figure 1.6: The board construction of the two finite differences. The circled terms cancel, leaving only the initial–final mismatch.

Lecture frame: 00:30:15.

Question from the audience. Does the vector fail to return to the same point, or does it return to the point with a different orientation? ^a

Response. The path closes at the same point. It is the transported vector that returns changed. The small separation drawn between V_A and V'_A is a picture of that change in the vector space at A .

^aLecture timestamp: 00:24:03–00:24:15.

The loop mismatch can be isolated by an intentionally engineered difference of differences:

$$\begin{aligned} & [(V_C - V_D) - (V_B - V_A)] \\ & - [(V_C - V_B) - (V_D - V'_A)] = V_A - V'_A. \end{aligned} \tag{1.12}$$

Every intermediate value cancels. The first bracket compares horizontal differences after a vertical displacement; the second compares vertical differences after a horizontal displacement. In the infinitesimal limit these become two second-order operations taken in opposite orders.

Question from the audience. Why subtract the two differences in this order rather than the reverse order? ^a

Response. The order fixes an orientation and is chosen so the intermediate vectors cancel. Reversing the order reverses the sign of the answer. Nothing physical depends on that choice once the loop orientation and the commutator convention are stated consistently.

^aLecture timestamp: 00:27:41–00:29:05.

Because the vectors at neighboring points must be compared by parallel transport, ordinary derivatives are not adequate. The finite differences become covariant derivatives. With the convention

$$[\nabla_\mu, \nabla_\nu]V^\alpha = R^\alpha{}_{\beta\mu\nu}V^\beta, \quad (1.13)$$

the leading holonomy around the oriented loop is

$$\Delta V^\alpha = R^\alpha{}_{\beta\mu\nu}V^\beta \epsilon^\mu \eta^\nu + O(\ell^3), \quad (1.14)$$

where $\Delta V^\alpha = V_A'^\alpha - V_A^\alpha$ for the corresponding orientation. The characteristic edge length is ℓ , so the leading change is of order ℓ^2 .

Only the antisymmetric part of the edge product contributes. Defining the oriented area bivector

$$\Sigma^{\mu\nu} = \epsilon^\mu \eta^\nu - \epsilon^\nu \eta^\mu, \quad (1.15)$$

we may equivalently write

$$\Delta V^\alpha = \frac{1}{2}R^\alpha{}_{\beta\mu\nu}V^\beta \Sigma^{\mu\nu} + O(\ell^3). \quad (1.16)$$

Question from the audience. As the two edge vectors go to zero, what else goes to zero? The Riemann tensor itself has no loop size in it. ^a

Response. The vector change goes to zero. Curvature is the local coefficient of that change per oriented area. Thus $\Delta V = O(\ell^2)$, while the Riemann tensor approaches a finite value at the base point. A finite loop can be understood, with suitable path ordering, as accumulated infinitesimal holonomy.

^aLecture timestamp: 00:50:50–00:52:24.

Question from the audience. Should we divide the long equation by the two edge lengths and call the result a mixed second derivative? ^a

Response. Dividing by area does isolate the curvature coefficient, but the geometric operation is not the mixed derivative of one scalar function. It is the noncommutativity of two covariant derivatives acting on a vector.

^aLecture timestamp: 00:58:23–00:58:39.

1.4 Expanding the Covariant-Derivative Commutator

Insert Eq. (1.7) into the commutator:

$$\begin{aligned} [\nabla_\mu, \nabla_\nu] &= [\partial_\mu + \Gamma_\mu, \partial_\nu + \Gamma_\nu] \\ &= [\partial_\mu, \partial_\nu] + [\partial_\mu, \Gamma_\nu] - [\partial_\nu, \Gamma_\mu] + [\Gamma_\mu, \Gamma_\nu]. \end{aligned} \quad (1.17)$$

The image shows a whiteboard with handwritten mathematical expressions. The top section contains two terms: $\partial x^\nu / \partial x^\mu \nabla_\nu \nabla_\mu V$ and $-\partial x^\nu / \partial x^\mu \nabla_\mu \nabla_\nu V$, followed by an equals sign and the expression $V_a - V_{a'}$. The bottom section shows the equation $\delta V = dX^\mu \delta x^\nu [\nabla_\nu, \nabla_\mu] V$.

$$\begin{aligned} & \partial x^\nu / \partial x^\mu \nabla_\nu \nabla_\mu V \\ & - \partial x^\nu / \partial x^\mu \nabla_\mu \nabla_\nu V \\ & = V_a - V_{a'} \end{aligned}$$

$$\delta V = dX^\mu \delta x^\nu [\nabla_\nu, \nabla_\mu] V$$

Figure 1.7: The difference of covariant differences compressed into a commutator. The two edge vectors supply the oriented area, and the commutator acts on the transported vector.

Lecture frame: 00:36:45.

$$\begin{aligned}
 & \partial_\nu \partial_\mu V^\alpha - \partial_\mu \partial_\nu V^\alpha + \Gamma_{\nu\mu}^\alpha V^\beta - \Gamma_{\mu\nu}^\alpha V^\beta \\
 & (\partial_\nu + \Gamma_{\nu}^\alpha) (\partial_\mu + \Gamma_{\mu}^\alpha) V^\beta - (\partial_\mu + \Gamma_{\mu}^\alpha) (\partial_\nu + \Gamma_{\nu}^\alpha) V^\beta \\
 & = \Gamma_{\nu\mu}^\alpha V^\beta - \Gamma_{\mu\nu}^\alpha V^\beta \\
 & - [\partial_\mu \Gamma_{\nu}^\alpha] = - \frac{\partial \Gamma_{\nu}^\alpha}{\partial x^\mu} \\
 & + [\partial_\nu \Gamma_{\mu}^\alpha] = + \frac{\partial \Gamma_{\mu}^\alpha}{\partial x^\nu}
 \end{aligned}$$

Figure 1.8: The operator expansion on the board. Ordinary derivative terms cancel, derivative–connection commutators produce $\partial\Gamma$, and the connection matrices leave a genuine matrix commutator.

Lecture frame: 00:42:45.

For smooth fields in a coordinate basis,

$$[\partial_\mu, \partial_\nu] = 0. \quad (1.18)$$

The mixed terms are evaluated with Eq. (1.11), while the last term remains because matrix multiplication is generally noncommutative:

$$[\nabla_\mu, \nabla_\nu] = \partial_\mu \Gamma_\nu - \partial_\nu \Gamma_\mu + \Gamma_\mu \Gamma_\nu - \Gamma_\nu \Gamma_\mu. \quad (1.19)$$

Question from the audience. What is the derivative of Γ_ν if Γ_ν is a matrix? ^a

Response. It is the matrix obtained by differentiating each entry:

$$(\partial_\mu \Gamma_\nu)^\alpha_\beta = \partial_\mu \Gamma^\alpha_{\nu\beta}.$$

The matrix indices were suppressed only to make the operator algebra legible.

^aLecture timestamp: 00:42:54–00:43:24.

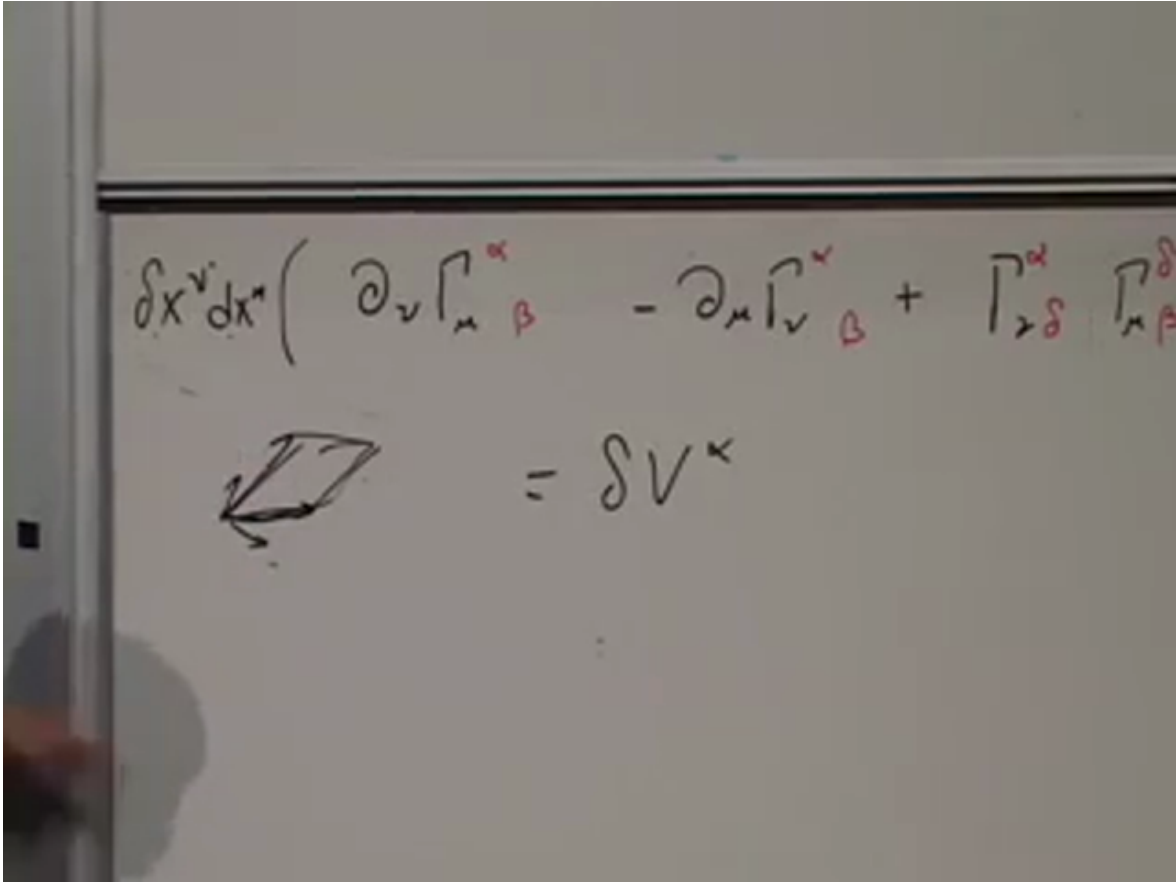


Figure 1.9: The completed board derivation: the connection derivatives and quadratic connection terms act on V^β , and the two edge vectors convert the curvature operator into the loop change ΔV^α .

Lecture frame: 00:47:15.

Restoring those indices gives the Riemann tensor in the convention of Eq. (1.13):

$$\boxed{R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\nu\beta} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\mu\beta}.} \quad (1.20)$$

Changing the order in the defining commutator changes the overall sign; changing the loop orientation does the same. The convention matters, but no geometric conclusion depends on selecting one convention rather than the other.

Question from the audience. Did choosing the two edge vectors select only one plane? Does the curvature depend on that choice? ^a

Response. The two vectors do select an infinitesimal plane, but they do not change the tensor. They contract the same $R^\alpha_{\beta\mu\nu}$ with a different area bivector and thereby select different components of the local curvature map.

^aLecture timestamp: 00:48:58–00:50:47.

The image shows a chalkboard with handwritten mathematical expressions. The top line is the Riemann curvature tensor formula with indices: $\partial_\nu \Gamma^\alpha_{\mu\beta} - \partial_\mu \Gamma^\alpha_{\nu\beta} + \Gamma^\alpha_{\nu\delta} \Gamma^\delta_{\mu\beta} - \Gamma^\alpha_{\mu\delta} \Gamma^\delta_{\nu\beta}$. Below this, an arrow points to the simplified expression: $R^\alpha_{\nu\mu\beta} = \partial_\nu \Gamma^\alpha_{\mu\beta} - \partial_\mu \Gamma^\alpha_{\nu\beta} + \Gamma^\alpha_{\nu\delta} \Gamma^\delta_{\mu\beta} - \Gamma^\alpha_{\mu\delta} \Gamma^\delta_{\nu\beta}$. Below that, the fully lowered tensor is written as $R_{\nu\mu\alpha\beta}$. The indices are color-coded: α is red, ν is blue, μ is green, and β is red.

Figure 1.10: The Riemann formula followed by its lowered-index symmetry bookkeeping. The board emphasizes sign reversal within either index pair and equality under exchange of the two pairs.

Lecture frame: 00:54:15.

1.5 Symmetries and the Content of Curvature

Lower the first index:

$$R_{\alpha\beta\mu\nu} = g_{\alpha\rho} R^\rho_{\beta\mu\nu}. \quad (1.21)$$

For the Levi-Civita connection, the fully lowered tensor satisfies

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \quad R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu}, \quad (1.22)$$

$$R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}, \quad R_{\alpha[\beta\mu\nu]} = 0. \quad (1.23)$$

The first antisymmetry follows because parallel transport preserves inner products when the connection is compatible with the metric. The second follows from the commutator. The third exchanges the two antisymmetric index pairs. The last is the algebraic, or first, Bianchi identity.

Question from the audience. Does the Riemann tensor have a cyclic or sum property? ^a

Response. Yes. In the present torsion-free geometry,

$$R_{\alpha\beta\mu\nu} + R_{\alpha\mu\nu\beta} + R_{\alpha\nu\beta\mu} = 0,$$

which is $R_{\alpha[\beta\mu\nu]} = 0$. Together with the pair symmetries, this identity removes additional apparent components.

^aLecture timestamp: 00:55:57–00:56:20.

These symmetries give

$$N_R(n) = \frac{n^2(n^2 - 1)}{12} \quad (1.24)$$

independent components in n dimensions. Thus the exact counts are

$$N_R(2) = 1, \quad N_R(3) = 6, \quad N_R(4) = 20.$$

A rough count of twenty-four is voiced during the lecture; the algebraic Bianchi identity supplies the missing reduction to twenty in four dimensions.

The derivative structure is equally important. Since $\Gamma \sim g^{-1}\partial g$, Eq. (1.20) contains

$$Riemann \sim \partial^2 g + (g^{-1}\partial g)(g^{-1}\partial g). \quad (1.25)$$

Some terms contain two derivatives acting on one metric component; others are quadratic in first derivatives. If the metric is known as a function of position, the curvature is therefore determined, although the component calculation can be laborious.

1.6 Ricci Tensor and Scalar Curvature

Contraction within either antisymmetric pair gives zero. For example,

$$g^{\mu\nu} R_{\alpha\beta\mu\nu} = 0, \quad (1.26)$$

because $g^{\mu\nu}$ is symmetric and $R_{\alpha\beta\mu\nu}$ is antisymmetric in μ, ν . The same argument applies to contraction of α, β .

The nontrivial mixed contraction is the Ricci tensor:

$$\boxed{R_{\beta\nu} = R^\alpha{}_{\beta\alpha\nu}.} \quad (1.27)$$

Equivalently, with all indices lowered,

$$R_{\beta\nu} = g^{\alpha\mu} R_{\alpha\beta\mu\nu}. \quad (1.28)$$

The symmetries in Eqs. (1.22)–(1.23) imply

$$R_{\mu\nu} = R_{\nu\mu}. \quad (1.29)$$

One further contraction gives the scalar curvature:

$$\boxed{R = g^{\mu\nu} R_{\mu\nu}.} \quad (1.30)$$

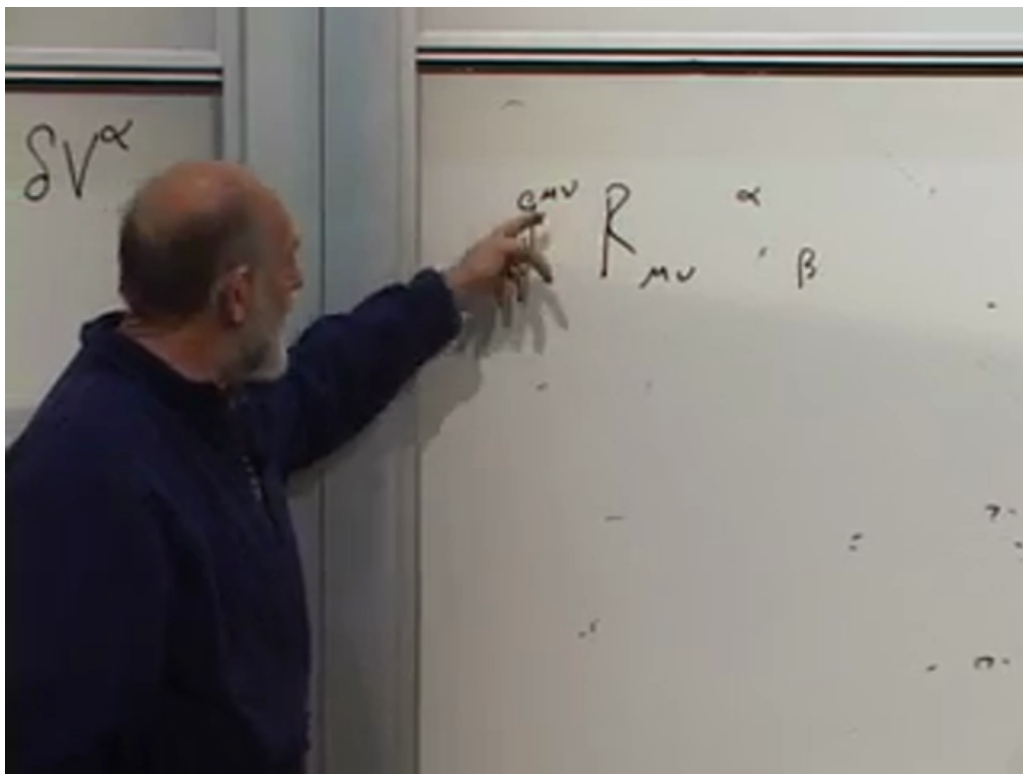


Figure 1.11: The board tests a symmetric metric contraction against an antisymmetric curvature pair. It vanishes, forcing the useful Ricci contraction to pair indices from different antisymmetric slots.

Lecture frame: 01:01:30.

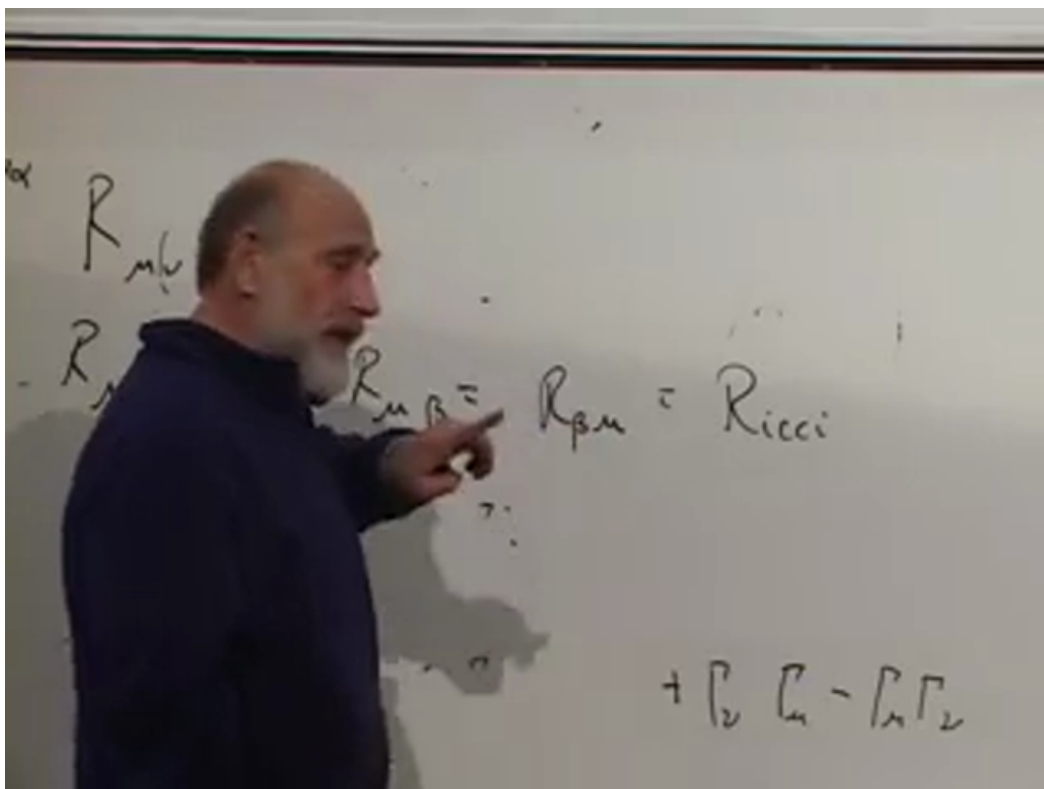


Figure 1.12: The mixed contraction producing the symmetric Ricci tensor, followed by the second contraction that produces the scalar curvature.

Lecture frame: 01:05:30.

Question from the audience. How exactly are two Riemann indices contracted? Is there a Kronecker delta hidden in the operation? ^a

Response. An upper index may be identified with a lower one and summed, as in $R^\alpha{}_{\beta\alpha\nu}$; this is equivalent to a Kronecker delta contraction. If all indices are lowered, use the inverse metric as in Eq. (1.28). Contractions within an antisymmetric pair vanish, while a mixed contraction gives Ricci, up to the sign fixed by the chosen Riemann convention.

^aLecture timestamp: 01:03:15–01:04:38.

Question from the audience. The same letter R is used for Riemann, Ricci, and scalar curvature. How can they be distinguished? ^a

Response. The indices carry the distinction: $R^\alpha{}_{\beta\mu\nu}$ is Riemann, $R_{\mu\nu}$ is Ricci, and R with no indices is the scalar. Writing the indices is not optional when the context could be ambiguous.

^aLecture timestamp: 01:05:00–01:05:28.

If $R \neq 0$ at a point, the geometry is certainly curved there. The converse is dimension-dependent. In two dimensions,

$$R_{\alpha\beta\mu\nu} = \frac{R}{2} (g_{\alpha\mu}g_{\beta\nu} - g_{\alpha\nu}g_{\beta\mu}), \quad (1.31)$$

so $R = 0$ everywhere is necessary and sufficient for local flatness. In three dimensions the Riemann tensor is completely determined by $R_{\mu\nu}$ and R , so $R_{\mu\nu} = 0$ implies flatness. In four or more dimensions a Ricci-flat geometry may retain Weyl curvature; the full Riemann tensor must vanish to guarantee local flatness.

Question from the audience. Would a saddle point be an example of zero scalar curvature on a curved two-dimensional surface? ^a

Response. No. A saddle has negative Gaussian curvature at the saddle point, and in two dimensions the scalar curvature is twice the Gaussian curvature. It is nonzero there. Cancellation among different curvature planes becomes possible only in higher dimensions.

^aLecture timestamp: 01:07:23–01:07:38.

1.6.1 Intrinsic curvature is not embedding curvature

A sheet of paper remains intrinsically flat when it is rolled into a cylinder. Distances and angles measured by an observer confined to the sheet have not changed, and small-loop parallel transport detects no intrinsic curvature. The rolled sheet does have extrinsic curvature because it bends inside the surrounding three-dimensional space. General relativity concerns the intrinsic geometry measured by observers living in spacetime, not how spacetime might be embedded in some larger arena.

Question from the audience. If a great circle is a geodesic on a sphere, does that make the surrounding space flat? ^a

Response. No. Geodesic is a property of a path; flatness is a property of a neighborhood tested by all sufficiently small loops. A sphere has nonzero intrinsic curvature even though its great circles are geodesics. A circle of fixed latitude, such as an Arctic circle away from the equator, is not a geodesic.

^aLecture timestamp: 01:08:21–01:09:20.

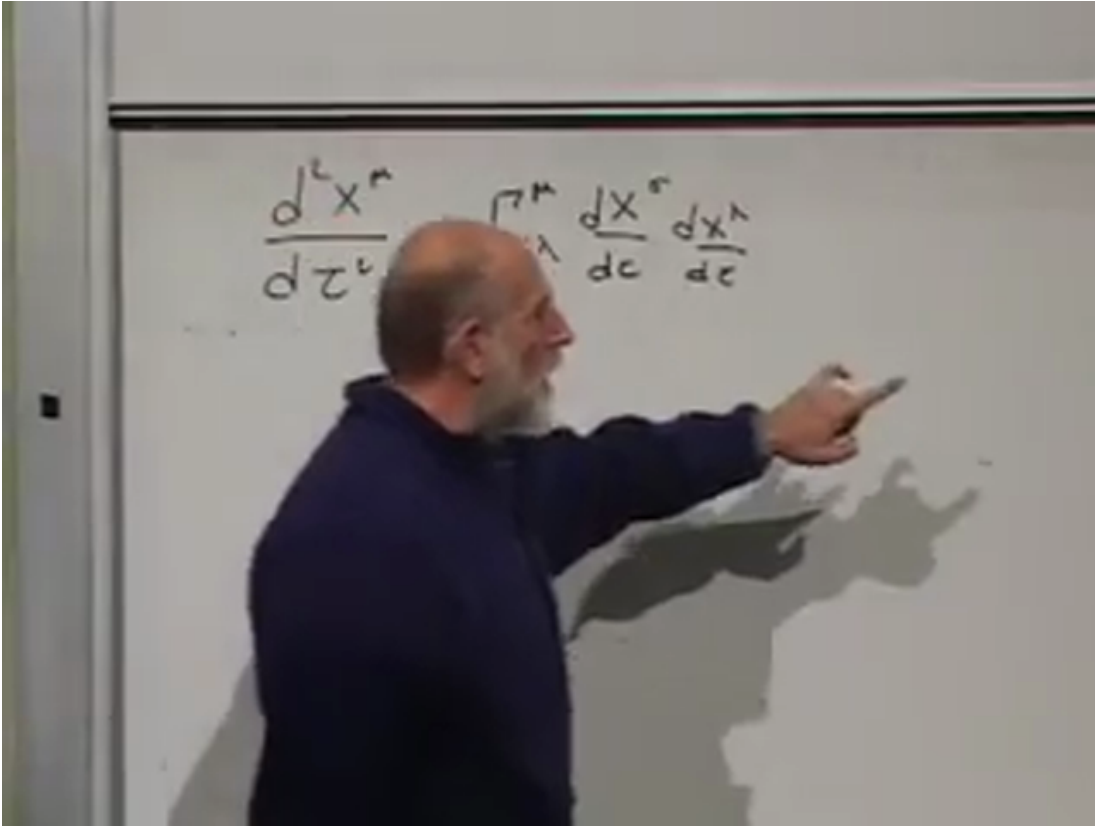


Figure 1.13: The geodesic equation, proper-time interval, and tangent vector on the board. The tangent is covariantly constant along a freely falling trajectory.

Lecture frame: 01:15:45.

1.7 Geodesics in the Newtonian Regime

We now turn from the geometry of curvature to the motion of matter. The relativistic rule is simpler than a force law: a freely falling particle follows a spacetime geodesic. If τ is proper time, the tangent vector is $u^\mu = dx^\mu/d\tau$, and parallel transport of the tangent along itself gives

$$\boxed{\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\sigma\lambda} \frac{dx^\sigma}{d\tau} \frac{dx^\lambda}{d\tau} = 0.} \quad (1.32)$$

With the $+- --$ signature and $c = 1$,

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (1.33)$$

for a timelike worldline.

To recover Newton, impose three linked approximations:

1. the particle velocities are small compared with the speed of light;
2. the metric is a small perturbation of Minkowski spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1;$$

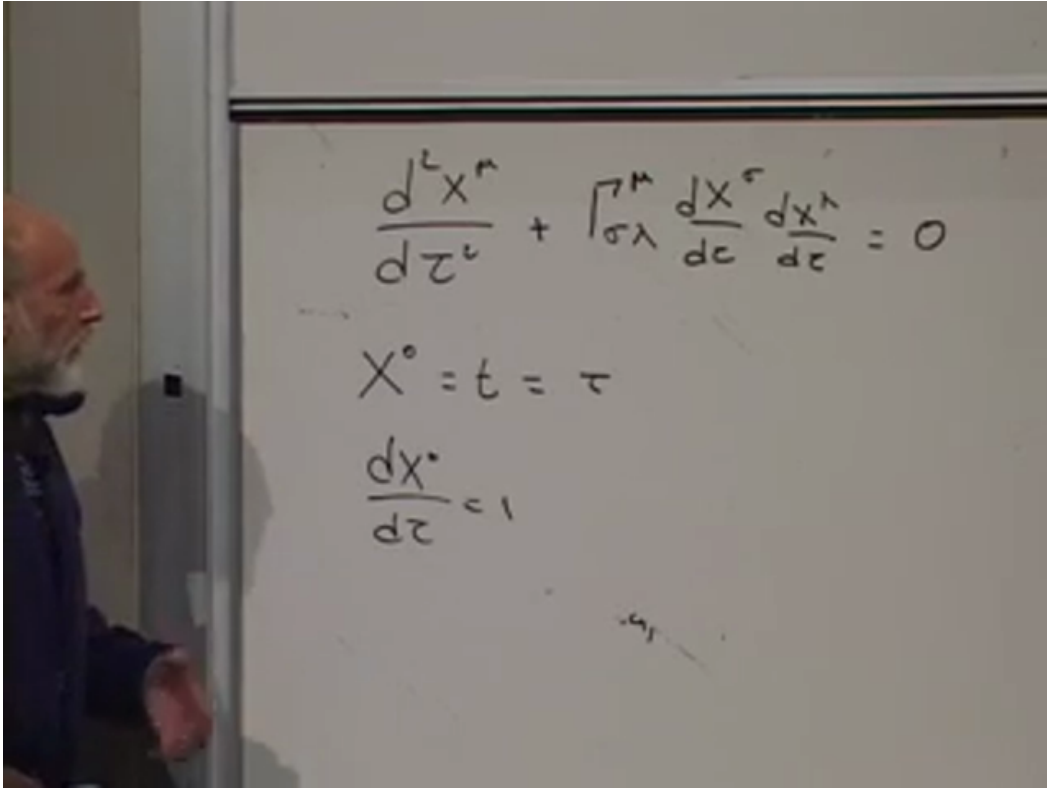


Figure 1.14: The slow-motion identification $x^0 = t \simeq \tau$ and $dx^0/d\tau \simeq 1$, displayed beside the full geodesic equation.

Lecture frame: 01:18:45.

3. the sources and metric vary slowly in time.

These assumptions describe planets and ordinary stars in a suitable frame. They fail near a black-hole horizon and are not a general description of an evolving universe.

For a slowly moving particle,

$$x^0 = t \simeq \tau, \quad \frac{dx^0}{d\tau} \simeq 1, \quad \frac{dx^i}{d\tau} \ll 1. \quad (1.34)$$

The second derivative of x^0 is correspondingly negligible at this order. The spatial components are the ones that retain the Newtonian acceleration.

In the product of two four-velocity components, the dominant term is therefore the 00 term. For a spatial index i ,

$$\frac{d^2 x^i}{dt^2} \simeq -\Gamma^i_{00}. \quad (1.35)$$

The connection is not itself a tensor, but in this chosen weak-field frame this component is the relativistic expression that reduces to the Newtonian acceleration field.

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\sigma\lambda}^\mu \frac{dx^\sigma}{d\tau} \frac{dx^\lambda}{d\tau} = 0$$

$$\frac{d^2 x^i}{dt^2} + \Gamma_{00}^i \frac{dx^0}{d\tau} \frac{dx^0}{d\tau} = 0$$

Figure 1.15: The spatial geodesic equation after retaining the dominant $dx^0/d\tau$ factors. The remaining connection component is Γ_{00}^i .

Lecture frame: 01:21:45.

Question from the audience. Why are both lower indices on the surviving Christoffel symbol zero? ^a

Response. Each lower index contracts with one component of the four-velocity. The spatial components $dx^i/d\tau$ are small, while $dx^0/d\tau \simeq 1$. Keeping the leading term therefore selects Γ^i_{00} .

^aLecture timestamp: 01:21:09–01:22:18.

1.8 The Time–Time Metric Component as Potential

Expand the relevant connection:

$$\Gamma^i_{00} = \frac{1}{2}g^{i\rho}(2\partial_0 g_{\rho 0} - \partial_\rho g_{00}). \quad (1.36)$$

The derivatives of the Minkowski part vanish, so every derivative in this expression is already first order in $h_{\mu\nu}$. Replacing the outside inverse metric by $\eta^{\mu\nu}$ is therefore consistent: including its $O(h)$ correction would produce only $O(h^2)$ terms.

The slowly varying source assumption makes the time derivatives $\partial_0 g_{\rho 0}$ negligible. Since $\eta^{ij} = -\delta^{ij}$, we obtain

$$\Gamma^i_{00} \simeq \frac{1}{2}\delta^{ij}\partial_j g_{00}, \quad (1.37)$$

and hence

$$\boxed{\frac{d^2 x^i}{dt^2} \simeq -\frac{1}{2}\delta^{ij}\partial_j g_{00}.} \quad (1.38)$$

The live sign bookkeeping pauses several times here, and rightly so: with signature $+ - - -$, losing one minus sign would turn attractive gravity into repulsive gravity. Keeping the metric convention explicit settles the result. Newtonian motion obeys

$$\frac{d^2 x^i}{dt^2} = -\delta^{ij}\partial_j \Phi. \quad (1.39)$$

Comparing Eqs. (1.38) and (1.39) gives

$$\partial_i g_{00} = 2\partial_i \Phi. \quad (1.40)$$

Integration determines the relation only up to a constant:

$$g_{00} = C + 2\Phi. \quad (1.41)$$

Far from an isolated source, $\Phi \rightarrow 0$ and the metric approaches Minkowski spacetime, so $C = 1$. Thus

$$\boxed{g_{00} \simeq 1 + 2\Phi} \quad (c = 1). \quad (1.42)$$

Restoring units gives $g_{00} \simeq 1 + 2\Phi/c^2$.

Question from the audience. Why does the comparison determine a constant and a factor of two rather than simply $g_{00} = \Phi$? ^a

Response. The acceleration comparison equates gradients, so an additive constant is invisible. The factor of two comes from the $\frac{1}{2}$ in the Christoffel symbol. The boundary condition $g_{00} \rightarrow 1$ where $\Phi \rightarrow 0$ fixes the constant.

^aLecture timestamp: 01:30:34–01:31:51.

The image shows a chalkboard with two equations written in black ink. The top equation is the geodesic equation for a particle moving in a gravitational field, written in terms of coordinate time τ and spatial coordinates x^μ :

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\sigma\lambda}^\mu \frac{dx^\sigma}{d\tau} \frac{dx^\lambda}{d\tau} = 0$$

To the right of this equation, the metric tensor $g_{\mu\nu}$ is written. The bottom equation shows the weak-field expansion of the Christoffel symbol Γ_{00}^i in terms of the metric components $g_{\alpha\beta}$ and their spatial derivatives:

$$\frac{d^2 x^i}{dt^2} = -\Gamma_{00}^i = -\frac{1}{2} \eta^{ik} \left(\frac{\partial g_{k0}}{\partial x^0} + \frac{\partial g_{0k}}{\partial x^0} - \frac{\partial g_{00}}{\partial x^k} \right)$$

Figure 1.16: The weak-field expansion of Γ^i_{00} . The connection contains two time derivatives of mixed metric components and one spatial derivative of g_{00} .

Lecture frame: 01:26:15.

The whiteboard contains three equations written in black marker:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\sigma\lambda}^\mu \frac{dx^\sigma}{d\tau} \frac{dx^\lambda}{d\tau} = 0$$

$$\frac{d^2 x}{dt^2} = -\Gamma_{00}^x = -\frac{1}{2} \left(\frac{\partial g_{x0}}{\partial x^0} + \frac{\partial g_{0x}}{\partial x^0} - \frac{\partial g_{00}}{\partial x} \right)$$

$$\frac{d^2 x}{dt^2} = -\frac{1}{2} \frac{\partial g_{00}}{\partial x}$$

Figure 1.17: The board comparison between geodesic acceleration and the gradient of the Newtonian potential. The boxed relation isolates the spatial derivative of g_{00} .

Lecture frame: 01:28:00.

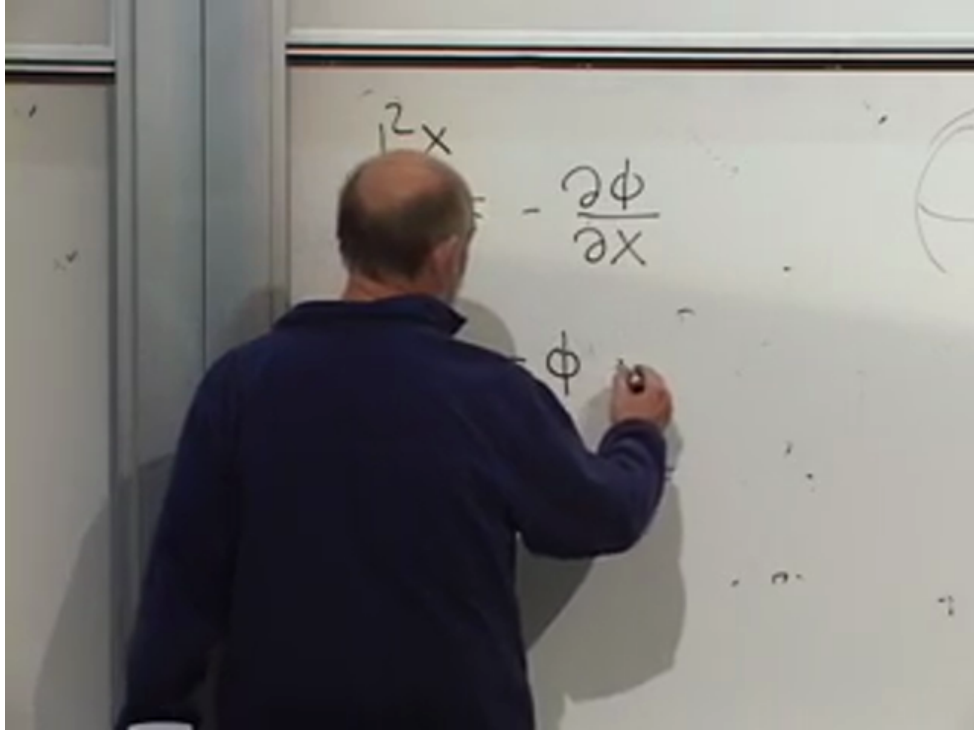


Figure 1.18: Newtonian acceleration above the integrated board relation between the time–time metric component and the potential. The additive constant is fixed by the asymptotically flat limit.

Lecture frame: 01:31:00.

1.9 Poisson’s Equation and the Source

Let $\mathbf{a}(\mathbf{x})$ be the Newtonian acceleration field. With the convention used above,

$$\mathbf{a} = -\nabla\Phi. \quad (1.43)$$

Gauss’s law for Newtonian gravity is

$$\nabla \cdot \mathbf{a} = -4\pi G\rho, \quad (1.44)$$

so the potential obeys Poisson’s equation:

$$\boxed{\nabla^2\Phi = 4\pi G\rho}, \quad \nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2. \quad (1.45)$$

In units $c = 1$, rest-mass density is the leading contribution to energy density. In the slow-moving frame,

$$T_{00} \simeq T^{00} \simeq \rho. \quad (1.46)$$

Poisson’s equation can therefore be written

$$\nabla^2\Phi = 4\pi GT_{00}. \quad (1.47)$$

Using $g_{00} = 1 + 2\Phi$, the constant disappears under the Laplacian:

$$\boxed{\nabla^2 g_{00} = 8\pi GT_{00}}. \quad (1.48)$$

We have learned that a metric component is controlled by an energy–momentum component. This is already the seed of the gravitational field equation.

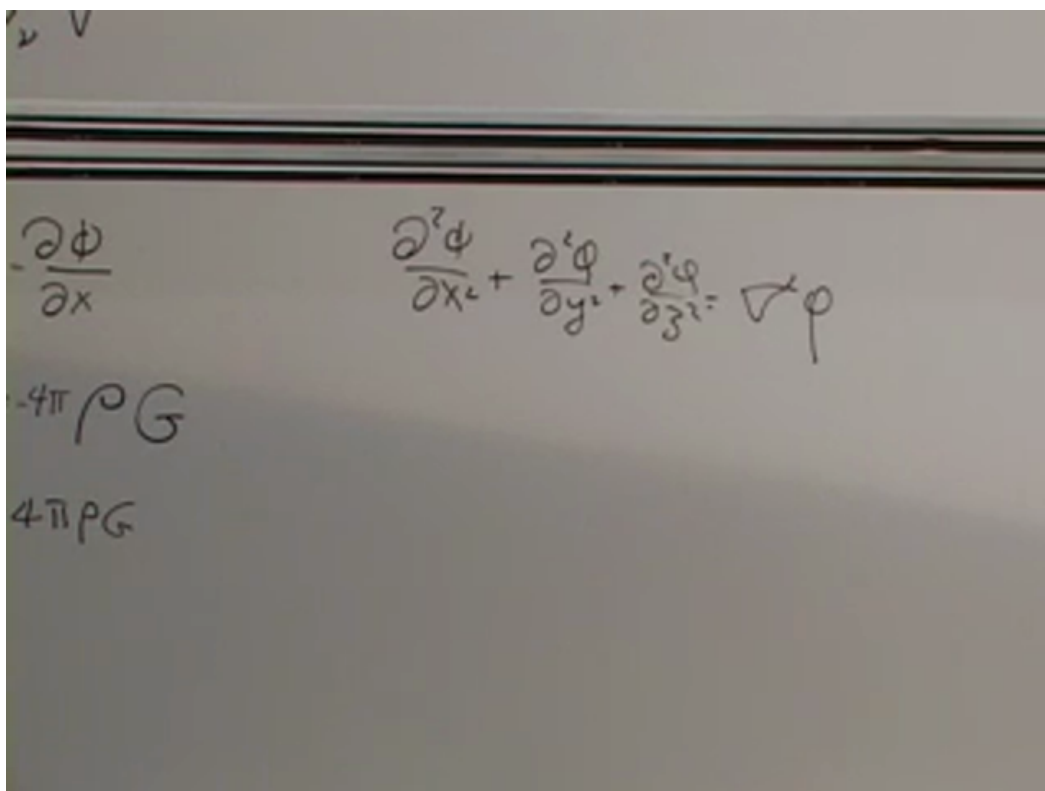


Figure 1.19: The Newtonian chain on the board: acceleration as a potential gradient, the divergence law, and Poisson's equation with its three-dimensional Laplacian.

Lecture frame: 01:35:30.

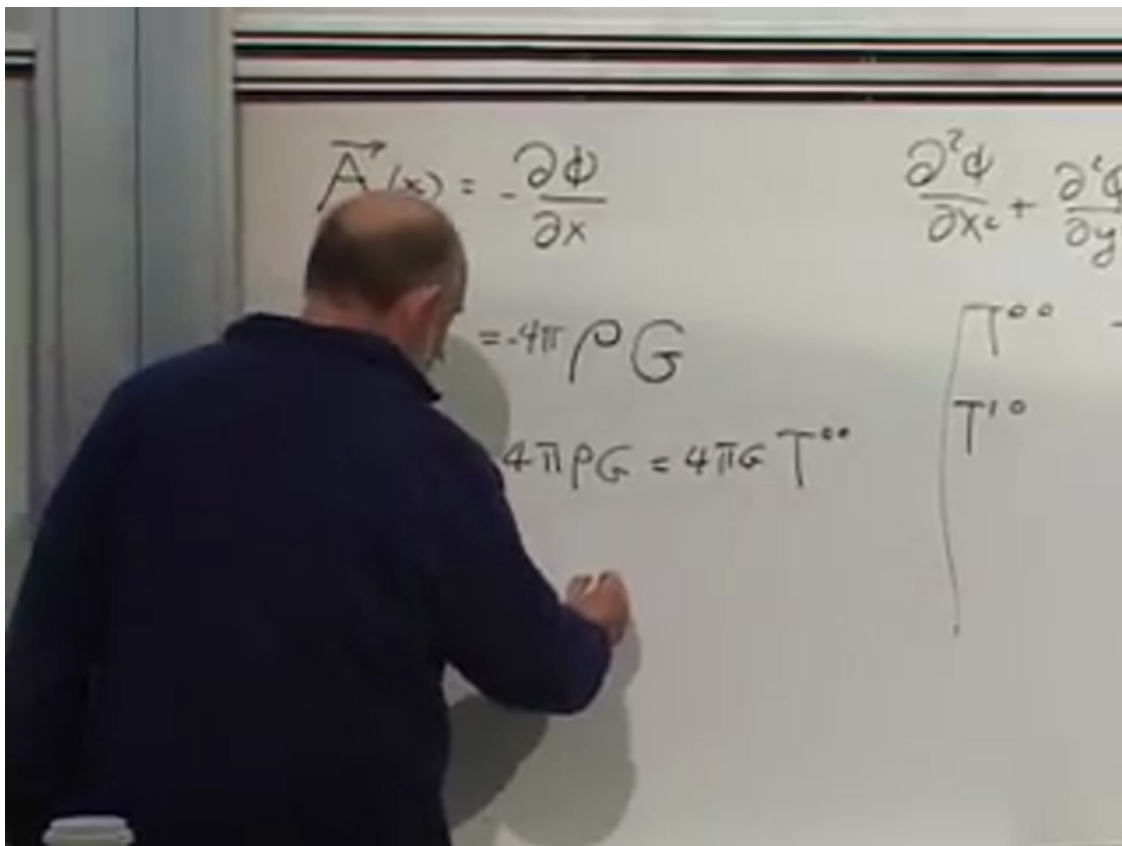


Figure 1.20: Poisson's equation rewritten with the 00 entry of the stress-energy tensor, followed by the weak-field equation for g_{00} .

Lecture frame: 01:37:45.

1.10 From One Component to a Tensor Equation

Equation (1.48) is useful but not yet generally covariant. It was derived in a special frame where velocities are small, the field is weak, and time dependence is slow. The right-hand side is one component of a tensor, while the left-hand side is written as a three-dimensional Laplacian of one selected metric component. A fundamental equation must instead equate complete tensors.

Question from the audience. What is wrong with $\nabla^2 g_{00} = 8\pi G T_{00}$ if it gives the correct Newtonian limit? ^a

Response. Nothing is wrong with it inside the approximation in which it was derived. It is incomplete because it is not manifestly a tensor equation and singles out one frame and one component. We seek a covariant equation whose 00 component reduces to it in the Newtonian regime.

^aLecture timestamp: 01:38:08–01:39:40.

Call the desired geometric tensor $G_{\mu\nu}$. It must

1. have two indices, like $T_{\mu\nu}$;
2. be built from the metric;
3. contain the second-derivative curvature structure; and
4. reproduce Eq. (1.48) in the Newtonian limit.

The target equation is

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (1.49)$$

The available symmetric two-index curvature tensors are $R_{\mu\nu}$ and $g_{\mu\nu}R$. Under the assumptions being used in the lecture, the candidate is therefore

$$AR_{\mu\nu} + Bg_{\mu\nu}R = 8\pi G T_{\mu\nu}, \quad (1.50)$$

where A and B are numerical coefficients. An overall normalization can be absorbed into Newton's constant; the nontrivial question is the relative coefficient.

The lecture stops before deriving the coefficients, but it reveals the destination:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R. \quad (1.51)$$

Why precisely this combination is selected, how its divergence relates to local energy–momentum conservation, and how its normalization recovers Newtonian gravity are the next steps. For now the logic is already visible:

$$\begin{aligned} \text{geodesics} &\longrightarrow g_{00} \simeq 1 + 2\Phi \\ &\longrightarrow \nabla^2 \Phi = 4\pi G \rho \\ &\longrightarrow G_{\mu\nu} = 8\pi G T_{\mu\nu}. \end{aligned}$$

Question from the audience. Are the coefficients difficult numbers? ^a

Response. No. With the Ricci term normalized to one, the second coefficient is $-\frac{1}{2}$. The simplicity of the numbers is not the reason for the combination; the geometric conservation identity is.

^aLecture timestamp: 01:45:13–01:45:33.

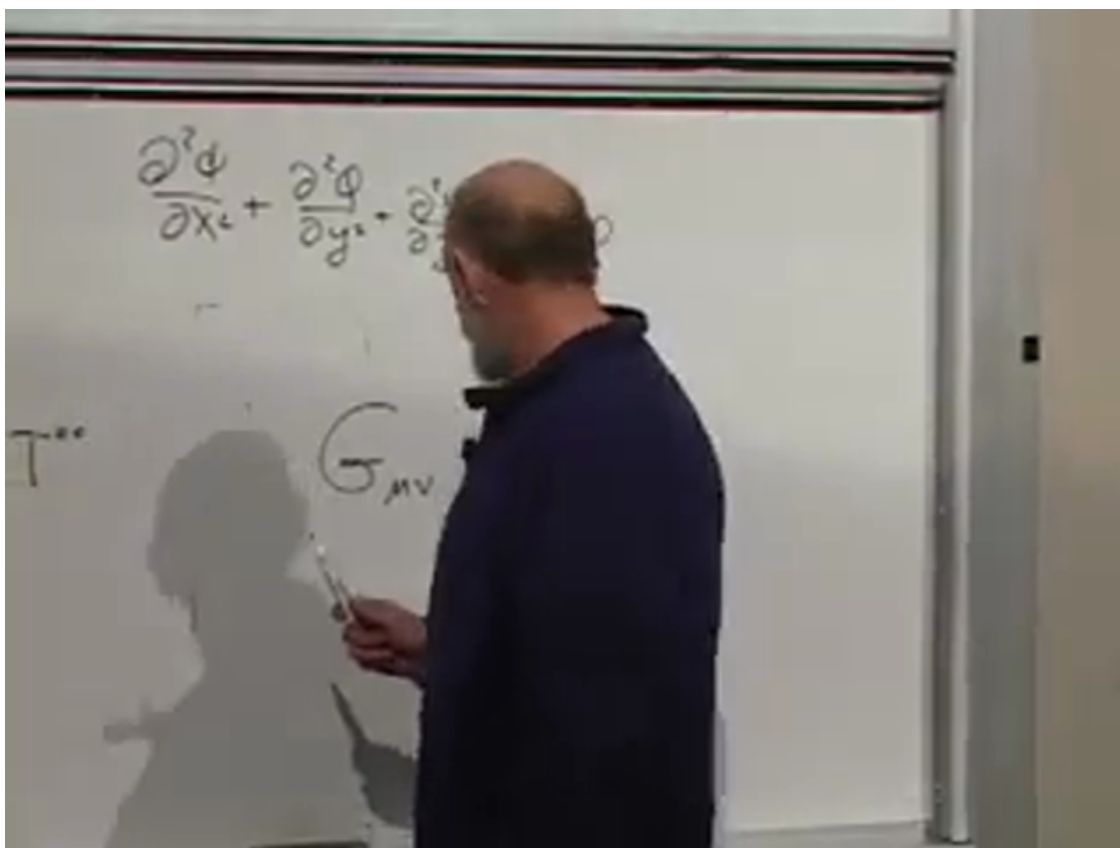


Figure 1.21: The weak-field component equation beside the proposed covariant form $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. The unknown left-hand side must be made entirely from the metric and curvature.

Lecture frame: 01:40:30.

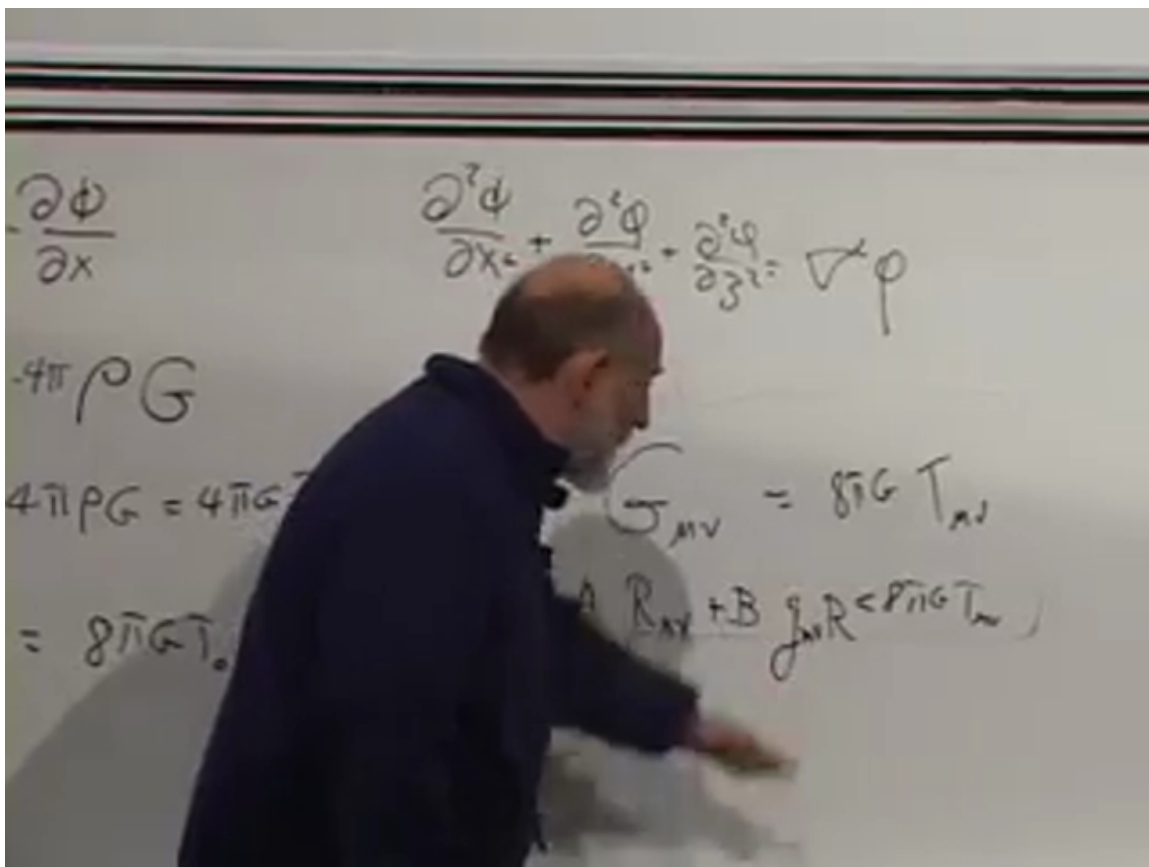


Figure 1.22: The final board ansatz: a linear combination of the Ricci tensor and $g_{\mu\nu}R$ is matched to the stress-energy tensor.

Lecture frame: 01:43:30.

1.11 What We Have Established

The operator notation $\nabla_\mu = \partial_\mu + \Gamma_\mu$ turns the geometric experiment of parallel transport into a calculation. A difference of differences around an infinitesimal loop is a commutator of covariant derivatives, and the resulting operator is the Riemann tensor. Its pair antisymmetries, pair exchange, and Bianchi identity determine its component structure; its mixed contractions produce the Ricci tensor and scalar curvature.

The same lecture then places the geometry under physical pressure. A slowly moving particle in a weak, nearly static field follows a geodesic whose spatial acceleration is $-\Gamma^i_{00}$. With signature $+ - - -$, this identifies $g_{00} \simeq 1 + 2\Phi$. Newtonian Poisson theory then becomes the component relation $\nabla^2 g_{00} = 8\pi G T_{00}$.

That component equation cannot be the final theory because it belongs to one approximation and one frame. The lecture therefore ends with a sharply posed reconstruction problem: find a symmetric two-index curvature tensor whose Newtonian limit is the component equation and whose covariant form can stand opposite $T_{\mu\nu}$. The only available combination is poised to become the Einstein tensor.